# Deprescribing Proton Pump Inhibitors in Children. Rebound Acid Hypersecretion, Tapering, Prescription Audit and the Conversation with Families.

Imagine a fourteen-month-old girl who comes to clinic for a routine visit. She is growing well, eating a varied diet and walking. She has been taking omeprazole since she was six weeks old, when she cried a lot and spat up after feeds. The prescription was started in the emergency department, and her family doctor has renewed it every month since. Her parents are worried about stopping, because they remember those difficult weeks. Nobody has ever written down why she is on it, or when it should end. Who should stop the omeprazole, how should it be done, and what should the family be told? This chapter is about those questions.

The main idea can be stated simply. Most children who take a proton pump inhibitor were started empirically, for symptoms that would have settled anyway, and most of those who stay on it do so because no one wrote a stop date. Stopping is safe for the great majority. But after a proton pump inhibitor ends, the stomach overshoots and makes more acid than before for a week or two, and this rebound is easily mistaken for a relapse of the original problem. A good deprescribing plan therefore names the indication, steps the dose down predictably, warns the family about the rebound in advance, offers a rescue option for that window, and arranges follow-up.

A few terms first. A proton pump inhibitor, which we will often simply call a PPI, is a drug such as omeprazole, esomeprazole, lansoprazole or pantoprazole that blocks the acid pump in the parietal cells of the stomach. An H2-receptor antagonist, or H2 blocker, such as famotidine, reduces acid by blocking the histamine signal to the parietal cell. Deprescribing means the planned and supervised reduction or stopping of a medicine that is no longer needed or whose harms may outweigh its benefits. Rebound acid hypersecretion means the temporary increase in acid production above the original level after a PPI is stopped. Gastro-oesophageal reflux, or GER, is the normal passage of stomach contents into the oesophagus, which is extremely common in infants. Gastro-oesophageal reflux disease, or GERD, is reflux that causes troublesome symptoms or complications such as oesophagitis.

The sources for this chapter include the healthy volunteer study by Reimer and colleagues in Gastroenterology in 2009 and a similar study by Niklasson and colleagues in 2010; a 2024 review of rebound acid hypersecretion by Namikawa and Björnsson; the Canadian evidence-based deprescribing guideline by Farrell and colleagues in 2017; a Cochrane review of deprescribing in adults by Boghossian and colleagues in 2017; the American Gastroenterological Association clinical practice update on deprescribing, known as the AGA update, by Targownik and colleagues in 2022; the American College of Gastroenterology guideline on GERD from 2022; the paediatric reflux guideline of the North American and European paediatric gastroenterology societies, NASPGHAN and ESPGHAN, by Rosen and

colleagues in 2018; the United Kingdom NICE guideline on reflux in children, known as NG1; infant trials of PPIs; Danish national prescribing data; quality improvement projects from the United States and Australia; and a study of how diagnostic labels shape parents' choices, by Scherer and colleagues in Pediatrics in 2013.

## Part one. Why the stomach overshoots.

To understand rebound, it helps to follow the feedback loop that controls acid secretion, step by step.

In the first step, the PPI binds irreversibly to the hydrogen potassium ATPase, the acid pump of the parietal cell, and the pH inside the stomach rises. In the second step, this higher pH removes a brake. Normally, acid in the antrum stimulates D cells to release somatostatin, which holds back the G cells that make gastrin. When the stomach is no longer acidic, somatostatin falls and the G cells release more gastrin. In the third step, gastrin acts as a growth signal. It stimulates the enterochromaffin-like cells, which release histamine and multiply, and it increases the mass of parietal cells. In effect, the stomach builds a larger acid factory while the PPI keeps the factory switched off. In the fourth step, the PPI is stopped. New pumps are made, and they now sit in an enlarged secretory apparatus driven by high gastrin and extra histamine. Acid output overshoots the original level. In the fifth step, recovery happens. Gastrin falls within days, but the extra cell mass takes weeks to months to regress. A marker called chromogranin A rises in parallel with the growth of enterochromaffin-like cells and falls as they regress.

This chain has an important consequence. Rebound is a predictable result of a drug that worked, not proof that the drug was needed. The longer and stronger the acid suppression, the larger the gastrin drive and the longer the overshoot. So symptoms in the first two weeks after stopping tell us very little about whether the original indication was real.

## Part two. Rebound in healthy people.

The clearest demonstration comes from a Danish study by Reimer and colleagues, published in 2009. They recruited 120 healthy volunteers with no history of reflux symptoms. Half received esomeprazole 40 milligrams a day for eight weeks, followed by four weeks of placebo. The other half received placebo for all twelve weeks. Nobody knew which group they were in. In the four weeks after the active drug was replaced by placebo, 44 percent of the former PPI group reported at least one acid-related symptom, such as heartburn, regurgitation or dyspepsia, compared with 15 percent of the group that had always taken placebo. The difference was highly significant. Looked at week by week, about 22 percent of the former PPI group had symptoms in week ten against 7 percent of the placebo group, and 21 percent against 2 percent in week twelve.

A similar Swedish study by Niklasson and colleagues in 2010 gave pantoprazole 40 milligrams or placebo for four weeks to 48 volunteers. After stopping, dyspepsia occurred in 44 percent of

the pantoprazole group compared with 9 percent of the placebo group, and the symptoms faded over the following three weeks.

The message is striking: people who never had reflux developed reflux symptoms simply because a PPI was stopped. In a patient, exactly the same symptoms would be interpreted as relapse unless the clinician predicts them. It is fair to add that an earlier systematic review, by Hunfeld and colleagues in 2007, found the older and smaller studies inconsistent, and that there is no published study of rebound in children. We therefore extrapolate from adults, which is reasonable for the physiology but uncertain for the size and duration of the effect.

## Part three. The time course.

The time course gives the clinician a script. According to the 2024 review by Namikawa and Björnsson, rebound symptoms usually begin between five and fourteen days after the last dose and, in the volunteer studies, last on average four to five days. Gastrin and chromogranin A return to normal within two to fourteen days after short courses. After more than a year of PPI therapy, however, hypersecretion has been shown to last longer than eight weeks, though less than twenty-six weeks. The AGA update notes that the hyperplasia of parietal and enterochromaffin-like cells may take two to six months to regress, and advises that symptoms persisting beyond about two months after stopping suggest either a true ongoing indication or a cause that is not acid.

In practice, this means three things. Tell the family that a flare is likely in the second week and that it should settle within days. Offer a rescue option for that window. And judge success at four to twelve weeks, not at day ten. Distinguishing rebound from the return of genuine reflux disease is difficult, which is why a planned review, rather than a phone call that restarts the drug, is the safeguard.

## Part four. Why deprescribe in children at all.

The strongest reason is that, in infants, the drug rarely did the work. The landmark trial by Orenstein and colleagues, published in 2009, randomised 162 infants with symptoms attributed to reflux, such as crying, arching and regurgitation, to four weeks of lansoprazole or placebo. Exactly the same proportion responded in each group: 54 percent. Serious adverse events, mostly lower respiratory tract infections, were more frequent with lansoprazole, ten against two. Two randomised withdrawal trials, in which all infants started on the drug and then half were switched to placebo, tell the same story. In the esomeprazole trial by Winter and colleagues in 2012, about 49 percent of infants switched to placebo stopped early because their symptoms worsened, compared with about 39 percent of those who stayed on esomeprazole, a difference that was not significant. In the rabeprazole trial by Hussain and colleagues in 2014, with 268 randomised infants, there was no significant difference from placebo on regurgitation, symptom scores or weight. The NASPGHAN and ESPGHAN guideline concludes that acid suppression is not needed for infants who only cry or fuss without warning

signs. In other words, stopping removes potential risk without removing benefit.

The second reason is the accumulating signal of harm, all from observational studies. In a United States military health system cohort of 792,130 children studied by Mitre and colleagues in 2018, PPI use in the first six months of life was associated with more than two and a half times the risk of food allergy, with an adjusted hazard ratio of 2.59, and with more anaphylaxis, with a hazard ratio of 1.45. A Taiwanese study by Wang and colleagues in 2021 found a hazard ratio for asthma of 1.57. A French nationwide study of more than 1.26 million children by Lassalle and colleagues in 2023 found a hazard ratio for serious infections of 1.34, higher for digestive and ear, nose and throat infections. And in the cohort studied by Malchodi and colleagues in 2019, PPI use in the first year was associated with a hazard ratio for fracture of 1.23, rising to 1.41 when PPIs were given for more than 150 days. None of these proves harm in an individual child, and confounding by indication is possible, because sicker children receive more drugs. But they all point in the same direction, and several rise with duration, which is the one exposure we control. The AGA update makes a balancing point: when there is a real indication, fear of an adverse event alone is not a reason to stop.

## Part five. Easy to start, hard to stop.

Population data show how prescriptions drift. In Denmark, Aznar-Lou and colleagues found that annual paediatric PPI use rose eight-fold between 2000 and 2015, from 0.1 to 3.1 users per thousand children. Sixty percent of children filled only one prescription, but 14 percent of the youngest children were still covered twelve months after starting. In Ireland, infant prescriptions rose from 1,011 in 2009 to 2,763 in 2018, almost a tripling. In 43 American children's hospital neonatal units between 2006 and 2013, almost one in four infants, 23.8 percent, received an H2 blocker or a PPI. And an audit of 400 hospitalised children in South Africa, published in 2026, found that 48 percent of PPI prescriptions were for indications not recommended by guidelines, while deprescribing had been attempted in fewer than a quarter. So most children receive a single short course, but a meaningful minority slide into long-term use without review. Hospital discharge, especially from neonatal units, is the commonest point of entry, and the first repeat prescription in primary care is where the end date disappears.

## Part six. Who should not stop.

Deprescribing does not mean stopping every PPI. The task is to separate a true indication from a habit. Children who should continue, with regular review, include those with severe erosive oesophagitis, meaning Los Angeles grade C or D, or a peptic stricture; Barrett's oesophagus; children after repair of oesophageal atresia who have documented oesophagitis; children with eosinophilic oesophagitis who are in histological remission on a PPI, because for them the PPI is the treatment; neurologically impaired children with proven oesophagitis or bleeding; children with ongoing high risk of upper gastrointestinal bleeding; and children in the middle of Helicobacter pylori eradication. Children who should be offered a trial off include infants with regurgitation, crying or arching without oesophagitis; children on an empirical course

started without a diagnosis whose symptoms have resolved; children who have completed a four- to eight-week course for typical heartburn; children in whom the only reason was a brief resolved unexplained event, apnoea, cough, hoarseness or so-called silent reflux; and children started on stress ulcer prophylaxis in intensive care who have now gone home. The bottom line is that a documented, ongoing indication is the only reason to continue.

## Part seven. What the adult trials tell us about strategies.

There is no paediatric trial comparing ways of stopping, so we borrow from adults. The Canadian guideline by Farrell and colleagues pooled trials in adults whose mild to moderate reflux had resolved, and compared the risk of symptom relapse with continuing the full dose. Stepping down to a lower daily dose barely increased relapse, with a risk ratio of 1.16, which was not significant. Switching to on-demand use, meaning taking the drug only when symptoms occur, increased relapse, with a risk ratio of 1.71. Stepping down to an H2 blocker increased it most, with a risk ratio of 1.92. The Cochrane review found that on-demand use saved almost four pills a week, at the cost of more symptoms and lower satisfaction. A systematic review by Haastrup and colleagues in 2014 found that between 14 and 64 percent of patients stopped successfully without worse symptoms, and that tapering seemed more effective than abrupt discontinuation. However, neither the Canadian guideline nor the AGA update found evidence that one particular taper schedule is better than another, and the AGA update accepts either tapering or abrupt discontinuation. The Canadian guideline gave a strong recommendation, based on low-quality evidence, to reduce the dose or stop and use on demand in adults whose symptoms have resolved after at least four weeks of treatment.

## Part eight. A practical taper.

Putting this together gives a pragmatic five-step plan for a child who has been on a daily PPI for more than about eight weeks. First, decide: confirm that there is no ongoing indication, document the reason for stopping and agree a review date. Second, if the child takes the drug twice daily, go to once daily, before breakfast, for one to two weeks. Third, give half the daily dose for one to two weeks. Fourth, give half the dose every second day for one to two weeks. Fifth, stop, with a rescue option available for the rebound window, and review at four and twelve weeks.

The plan should fit the child. A thriving infant on a short course can stop directly or over two to four weeks. The parent handout from the Royal Children's Hospital in Melbourne, for example, uses half the dose daily for two weeks, then half the dose every second day for a week, then stops. Children on years of high-dose therapy may need slower steps. Rescue options are an antacid or an occasional dose of an H2 blocker. Regular H2 blockers should be avoided as a long-term substitute, because tolerance develops within weeks; the American Academy of Pediatrics noted tachyphylaxis within about six weeks. The drug should be restarted only for symptoms that persist beyond the rebound window.

## Part nine. Different children, different goals.

The same drug calls for different plans. In the infant, regurgitation is physiology. The plan is to stop, with an optional short taper, and to offer feeding advice: thickening, sensible feed volumes, and a trial of an extensively hydrolysed formula if cow's milk protein allergy is suspected. In the school-age child, the question is whether the planned four- to eight-week course was completed; if so, taper over two to six weeks, and consider endoscopy if symptoms keep returning. In the adolescent, reflux behaves more like adult disease, on-demand use is acceptable, and lifestyle factors such as weight and late meals matter. The two hardest groups are neurologically impaired children and children after oesophageal atresia repair. In both, true oesophagitis is frequent and symptoms are hard to read, and after atresia repair there is a long-term risk of Barrett's oesophagus. Here, deprescribing should follow endoscopic findings rather than calendar dates.

## Part ten. What the guidelines say.

The guidelines agree on some points and leave gaps on others. On empirical PPIs in infants, they are aligned: the NASPGHAN and ESPGHAN guideline says not to use them for crying or fussing and not as a diagnostic test, and NICE says not to offer them for isolated regurgitation. On the length of a treatment trial they differ slightly: NASPGHAN and ESPGHAN suggest four to eight weeks, NICE suggests four weeks followed by assessment, and the American College of Gastroenterology, for adults, recommends trying to stop after a successful eight-week course. On reassessment they are aligned: NASPGHAN and ESPGHAN advise periodic attempts at weaning, NICE advises referral if symptoms recur after stopping, and the AGA advises regular review of the indication. But on how to taper, on warning patients about rebound, and on who should continue long term, the paediatric guidelines say nothing, and the advice comes from adult documents. The most material gap is the rebound warning: only the AGA update explicitly advises telling patients to expect transient symptoms. In short, paediatric guidelines say when to start and when to reassess, but not how to stop.

## Part eleven. Prescription audit and quality improvement.

Because no one else will do it, a unit can audit its own prescriptions with five questions for every PPI on the list. Is a diagnosis documented, or only a symptom? Has the planned four- to eight-week course already ended? Is the dose weight-appropriate, once daily and before a meal? Does the discharge letter carry a stop date and a review? And who owns the decision: the family doctor, the paediatrician or the gastroenterologist? The audit is most useful at discharge from neonatal and intensive care units, where many prescriptions start, and at the first repeat prescription in primary care, where they persist. A French tool called POPI, with 104 criteria for inappropriate prescribing and omissions in children, is one structured way to flag such prescriptions. The effect comes from feeding the results back to prescribers.

Quality improvement work shows that this changes practice. On an infant unit described by Angelidou and colleagues in 2017, a written guideline reduced prescriptions without an indication from about seven and a half per month to none, and all acid-suppression prescriptions from about eleven and a half to two and a half per month, with no change in balancing measures. Across three hospitals in Victoria, Australia, a project that included a parent script and handouts reduced acid-suppression prescriptions by 21 percent and increased the proportion of clinicians advising parents to stop by 46 percent.

## Part twelve. The conversation with families.

The hardest part of deprescribing is often the conversation, and here there is useful evidence. Scherer and colleagues, in a study of 175 parents published in Pediatrics in 2013, showed that labelling an infant's crying and spitting as a disease, GERD, increased parents' interest in medication, even when they were told that the medication was ineffective. Without the disease label, interest in medication appeared only when its lack of efficacy was not discussed. Families also remember the difficult weeks of crying, and a rebound flare seems to confirm their fear. And ownership is often unclear: a specialist starts the drug, the family doctor renews it, and nobody stops it.

So the words matter. Instead of saying that the child has reflux disease, say that spitting up is normal at this age and usually stops by twelve to fourteen months. Instead of saying that we will see how she goes off it, say that there may be a few worse days in the second week and that this will settle. Instead of saying to restart the drug if symptoms come back, say to use the rescue dose and to call at four weeks if symptoms persist. Instead of saying that these drugs are dangerous, which backfires when a drug is genuinely needed, say that the medicine is not helping now and that every medicine has some risk. And instead of saying that the family doctor will decide, say that you will write the plan and the stop date in the letter. In short: name the normal, predict the rebound, give a rescue and a date, and own the decision.

## Part thirteen. Traps and pearls.

Several traps are common. The first is concluding that the child needs the drug because symptoms returned; symptoms between days five and fourteen are rebound, and success should be judged at four to twelve weeks. The second is switching long term to an H2 blocker; tolerance develops within weeks, so H2 blockers are for rescue only. The third is stopping everyone; eosinophilic oesophagitis in remission, severe oesophagitis and Barrett's oesophagus are real indications. The fourth is stopping without explaining why, so that the family restarts the drug at the first flare. The fifth is assuming that no symptoms means no oesophagitis in a neurologically impaired child or a child after atresia repair; endoscopy should guide those decisions. And the last is deprescribing the PPI while leaving the cause untouched, such as cow's milk protein allergy or overfeeding in an infant, obesity or late meals in an adolescent, or an eosinophilic oesophagitis that was never biopsied. Stopping the drug is the end of a diagnostic process, not a substitute for one.

Let us return to a few scenarios. The thriving fourteen-month-old from the start of this chapter should stop over two to four weeks, with a warning about rebound, because she has no indication and the infant trials were negative. If her mother calls on day eight because the crying and spitting are back, the answer is a rescue dose, reassurance and keeping to the plan, because this is the rebound window. A twelve-year-old whose heartburn resolved after eight weeks can halve the dose and then stop, with on-demand use if needed. An adolescent with eosinophilic oesophagitis in histological remission on a PPI should continue, and should have repeat endoscopy before any change, because the PPI is the treatment. A child after oesophageal atresia repair with Los Angeles grade C oesophagitis should continue, with endoscopic surveillance. An infant leaving the neonatal unit on famotidine for reflux should have it stopped at discharge, with the decision written in the letter. And a child whose symptoms persist three months after stopping has moved beyond rebound, and needs reassessment with endoscopy, pH-impedance monitoring and attention to diet.

So, tomorrow in clinic, carry these points. After eight weeks of a PPI, 44 percent of healthy volunteers developed reflux symptoms on stopping, compared with 15 percent on placebo. Rebound starts between days five and fourteen and usually lasts days, although after long use hypersecretion can last eight to twenty-six weeks; symptoms beyond about two months are not rebound. In infants, lansoprazole and placebo each produced a 54 percent response, with more lower respiratory infections on the drug. Observational studies link early PPI use to food allergy, asthma, serious infection and fracture, with hazard ratios between about 1.2 and 2.6. In Denmark, paediatric PPI use rose eight-fold, and 14 percent of the youngest children were still on it a year later. In adults, lowering the dose barely increased relapse, whereas on-demand use and switching to an H2 blocker increased it more, and tapering worked better than an abrupt stop, although no schedule is proven best. Continue the drug for severe oesophagitis, stricture, Barrett's oesophagus, eosinophilic oesophagitis in remission and high bleeding risk. And remember that a disease label makes parents want medicine, even when they are told it does not work. Every prescription you write should carry an indication, an end date and the name of the person who will stop it.